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Industrial Systems Modernization | Manufacturing Data & Automation

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Signal Twin Review

A practical review sheet for deciding whether a legacy function is ready to move.

FUNCTION IDENTITY

Field	Review prompt
Business function	What production decision or action does this function enable?
Live users	Operators, supervisors, quality, maintenance, engineering, R&D, support?
Source signals	PLCs, devices, tags, manual inputs, schedules, recipes, external systems?
Logic	States, calculations, queries, scripts, transactions, exception handling?
Consumers	Displays, reports, SQL jobs, PI points/AF, alerts, data lake, external apps?
Consequence	What happens when the function is wrong, late, unavailable, or ambiguous?
Owner	Who accepts behavior, data, release, and ongoing support?

TARGET TWIN

Layer	Target definition
Ignition application	Project/component boundaries, reuse, navigation, state, diagnostics, security.
Tag/UDT model	Naming, types, metadata, alarms, quality, ownership, versioning.
SQL interaction	Named queries, transactions, schema, error handling, performance, access.
Historian context	Point mapping, asset hierarchy, units, timestamps, quality, retention, consumers.
Cloud contract	Schema, semantics, freshness, quality, access, owner, intended consumers.

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Release Decision

Promote, hold, repair, or retire based on explicit evidence.

ACCEPTANCE MATRIX

Check	Evidence required	Status
Function parity	Normal states, transitions, calculations, commands, exceptions	Pass / Hold / Repair
Data integrity	Names, values, units, timestamps, quality, completeness, reconciliation	Pass / Hold / Repair
Operator usability	Task flow, visibility, alarms, acknowledgment, recovery, training	Pass / Hold / Repair
Performance	Response, query/load behavior, store-forward, degradation handling	Pass / Hold / Repair
Security and access	Roles, credentials, least privilege, network path, auditability	Pass / Hold / Repair
Supportability	Diagnostics, logs, runbook, owner, escalation, source/version	Pass / Hold / Repair
Rollback	Trigger, authority, procedure, data reconciliation, communication	Pass / Hold / Repair
Observation	Window, measures, owner, incident path, learning capture	Pass / Hold / Repair

DECISION RECORD

Promote Evidence supports cutover; ownership and rollback are explicit.	Hold Evidence is incomplete or timing/production conditions are wrong.
Repair Target behavior, data context, usability, performance, or support needs change.	Retire The legacy function is not worth reproducing; remove with consumer approval.

LEARNING RETURN

What component, naming rule, test, diagnostic, runbook, data contract, or governance decision can the next modernization tranche reuse?